

Sadhvika Pachika | Data Engineer

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SUMMARY

Results-driven **Data Engineer** with over 4+ years of experience designing and implementing robust ETL pipelines, scalable cloud-based architectures, and data solutions that empower business insights and innovation. Expertise in **big data processing**, **data warehousing**, and **cloud platforms** such as AWS and Azure, with a strong focus on compliance and efficiency. Proficient in tools like **PySpark**, **PostgreSQL**, and **Apache Airflow**, with hands-on experience in handling complex healthcare and financial datasets, ensuring **HIPAA compliance** and maintaining **data security**. Adept at collaborating with cross-functional teams to deliver scalable, cost-effective, and real-time analytics platforms that drive data-driven decision-making. Proven ability to streamline operations, enhance data accessibility, and enable predictive and prescriptive modeling for improved business outcomes. Skilled in visualizing key metrics with tools like **Tableau** and **Power BI** to facilitate strategic insights for stakeholders.

TECHNICAL SKILLS

Programming Languages:	Python, SQL, R Programming
Libraries:	PySpark, pandas, NumPy, Scikit-learn
Cloud Platforms:	AWS (S3, Glue, Redshift, Lambda, EMR), Azure (Data Factory, Synapse Analytics)
Hadoop ecosystem :	HDFS, Spark, Hive, MapReduce, Linux OS, Cassandra, Putty, pig
Data Warehousing & Databases:	Amazon Redshift, PostgreSQL, MySQL, SAP ERP Integration
Data Visualization:	Tableau, Power BI, Excel
DevOps Tools :	Jenkins, Informatica, Docker, Ansible, Maven
Security & Compliance:	HIPAA-compliant data handling, encryption protocols, tokenization, differential privacy
Collaboration :	Git, Jira, Confluence, Agile scrum methodology
Data Processing :	Apache Spark, Hadoop, Kafka, Airflow

PROFESSIONAL EXPERIENCE

Anthem, CA Data Engineer	Aug 2024 - Current
- Designed and implemented a secure cloud-based data sandbox on AWS, enabling external collaborators to access de-identified healthcare datasets for AI and analytics innovations. - Enhanced ETL workflows using PySpark and AWS Glue, processing over 500,000 records daily with minimal latency and error rates. - Spearheaded the optimization of Amazon Redshift data warehouses, applying compression and partitioning techniques to improve query performance by 40%. - Developed robust Apache Airflow DAGs to orchestrate and monitor end-to-end workflows, introducing alerting mechanisms for pipeline failures and automated recovery. - Applied advanced privacy-preserving techniques, including tokenization and differential privacy, ensuring 100% compliance with HIPAA and other regulatory requirements. - Collaborated with cross-functional teams, including data scientists and analysts, to deliver real-time insights through Power BI dashboards, empowering data-driven decision-making. - Reduced cloud storage costs by 20% through Parquet compression and implementing lifecycle policies for archiving aged data in AWS Glacier. - Optimized complex SQL queries and implemented indexing strategies in Amazon Redshift, reducing query execution times by 30% and improving data retrieval efficiency for large-scale analytics workflows.	
Data Engineer Intern (Anthem)	Dec 2023 – May 2024
- Developed and automated ETL pipelines using Python, PySpark, and Apache Airflow, enabling seamless integration and processing of de-identified healthcare data from multiple sources. - Designed a multi-tier data storage architecture on AWS S3, organizing data into raw, processed, and analytics zones to improve data accessibility and scalability. - Built reusable data transformation workflows that standardized schemas to the FHIR standard, enhancing interoperability across datasets and systems. - Conducted real-time data validation and monitoring, ensuring compliance with HIPAA and maintaining detailed audit logs for all data processes. - Collaborated with data scientists to prepare curated datasets for machine learning models, accelerating the development of predictive analytics tools. - Partnered with business analysts to align ETL outputs with business objectives, contributing to dashboards in Power BI. - Wrote and optimized SQL queries to extract, transform, and load (ETL) large datasets from relational databases into AWS Redshift, improving query performance and reducing data processing time by 20%.	
Harman Connected Services, India Data Engineer	Dec 2020 – Aug 2022
- Designed and maintained ETL pipelines to collect, process, and store high-volume data from smart audio devices and connected home solutions, ensuring real-time data availability for analysis and machine learning workflows. - Collaborated with product and engineering teams to analyze user interaction data from JBL and Harman Kardon devices, providing actionable insights to enhance audio quality, connectivity, and overall user experience.	

- Built scalable **data models** to support personalized audio settings for consumers, optimizing sound quality and customization features across Harman products, including **wireless speakers and headphones**.
- Partnered with **data scientists** to deploy and monitor infrastructure for **machine learning models**, enabling predictive features such as **battery health monitoring, maintenance alerts, and adaptive sound settings**.
- Utilized **AWS (S3, Lambda, Redshift)** and **Python** for processing and managing data, reducing pipeline execution time by **20%** and ensuring scalable storage solutions.
- Developed dashboards in **Tableau** to visualize user engagement metrics and device performance trends, empowering product teams with data-driven insights for feature enhancements.
- Improved **data pipeline reliability** through robust testing and automation, achieving a **15% reduction in data processing errors** and enhancing system uptime.

Intex Technologies, India

Oct 2019 – Nov 2020

Data Engineer

- Developed and implemented real-time data integration pipelines using **Apache Kafka**, streamlining machine sensor data transmission to cloud systems and improving operational efficiency.
- Built and optimized ETL pipelines with **PySpark, Python, and SQL**, ensuring seamless data extraction, transformation, and loading into **Azure Synapse Analytics** and **Azure Data Lake**.
- Automated and scheduled data workflows with **Apache Airflow**, enhancing the efficiency of data processing tasks and reducing manual intervention by 30%.
- Collaborated with data scientists to prepare and transform machine sensor data for predictive maintenance models, resulting in a 15% reduction in unplanned downtime.
- Managed and optimized **Azure Synapse Analytics** cloud-based data warehouses, improving query performance and ensuring high data availability for business intelligence reporting.
- Developed business intelligence dashboards using **Power BI**, delivering real-time insights into machine performance, downtime trends, and maintenance schedules.
- Integrated legacy systems with cloud infrastructure via **RESTful APIs**, enabling seamless data flow and ensuring real-time decision-making capabilities across departments.

Data Analyst (Intex Technologies, India)

Jan 2019 – Sep 2019

- Consolidated and transformed data from **SAP ERP, CRM** systems, and inventory tools into a centralized repository using **SQL** and **Python**, improving data quality and accessibility by **30%**.
- Built and deployed a predictive sales forecasting model using **Python (Scikit-learn)**, increasing forecast accuracy by **20%** and reducing stockouts and overproduction.
- Designed interactive **Power BI** dashboards to visualize financial trends and supply chain **KPIs**, enabling leadership to make data-driven decisions and improve profitability.
- Automated data cleaning and validation pipelines using **Python (pandas, NumPy)**, reducing manual processing time by **25%** and ensuring data consistency for analytics.
- Deployed automated data pipelines on **Azure** Data Lake to store and manage large datasets, enhancing accessibility for cross-functional teams and reducing data retrieval times by **20%**.
- Conducted advanced profitability analysis using **Excel** to identify low-margin product lines and highlight cost-saving opportunities, contributing to a **7%** increase in profit margins.
- Collaborated with senior leadership and cross-functional teams to align on cost optimization strategies, presenting actionable insights through interactive dashboards and reports, leading to a **15%** reduction in operational costs.

PROJECTS

Orders Data Analysis

- **Designed and implemented robust data transformation workflows using dbt**, converting complex SQL queries into reusable models and thoroughly documenting data sources. Successfully integrated **customers, orders, and payments datasets from Amazon S3 into Snowflake**, ensuring seamless data ingestion and accessibility for analysis.
- **Established automated testing protocols within dbt** to validate data consistency, accuracy, and integrity across integrated datasets. These tests enhanced reliability for downstream analytics and empowered data teams with trustworthy insights.

Financial Market Analysis and Prediction

- **Data Integration and Visualization:** Extracted, cleaned, and merged data from multiple sources, ensuring accuracy and reliability. Leveraged Python, SQL, and Power BI to manipulate data and create interactive dashboards that highlighted critical business trends and insights.
- **Strategic Analysis and Reporting:** Conducted in-depth analysis to address complex business challenges, optimizing key performance indicators (KPIs) to enhance efficiency by 20%. Delivered comprehensive reports and actionable recommendations to stakeholders, driving informed decision-making.

EDUCATION & CERTIFICATION

Master of Science, Information systems and Technologies	May 2024
University of North Texas	
Bachelor of Science, Computer Science and Engineering	May 2020
Jawaharlal Nehru Technological University	
Dbt Certification: Dbt Fundamentals Snowflake Hands-On Essentials	